

Project Description and Business Rules

Academic Examination & Remote Proctoring System

This project focuses on designing a centralized database for a university's examination and remote proctoring operations. The system supports the core academic structure, tracking faculties, departments, programs, and course enrollments, but it is heavily specialized toward conducting secure assessments. It features a robust question bank, customizable exam generation, scheduling, and grading modules. To ensure academic integrity, a dedicated proctoring module captures student telemetry (keystrokes, window focus, clipboard activity) during exam attempts, automatically flags suspicious behavior through similarity comparisons, and integrates with a formal disciplinary review and appeals process.

Organizational and Academic Structure A university consists of multiple faculties, and each faculty contains several departments. Each department is responsible for offering various academic programs, and every student must be enrolled in one specific program. Departments also maintain the catalog of courses and employ instructors. A course can require other courses as prerequisites before a student can take it.

Course Offerings and Enrollment A university runs multiple semesters. During a semester, a department schedules specific courses as course offerings. Each course offering is taught by a designated instructor. Students enroll in these course offerings to complete their programs. If a course offering reaches its maximum capacity, students are placed on a waitlist. Upon completion of a course offering, the student receives a formal enrollment grade based on standardized grade scales and bands.

Question Banks and Exam Generation Instructors author and manage questions within department-specific question banks. These banks are organized into specific academic topics. Questions can have multiple options and file attachments. Instructors create exams for their assigned course offerings. An exam is structured into distinct sections, and each section contains a specific sequence of questions pulled from the question bank.

Exam Scheduling and Attempts Exams are scheduled for specific dates and times and are allocated to physical or virtual exam rooms. Instructors are assigned to these rooms to supervise the exams. Students must register for an exam to participate. When a student takes an exam, the system records this as an attempt. During an attempt, the student selects options or provides text answers to the exam questions. These answers are then graded by an instructor or graded automatically, producing a final calculated score and result for the attempt.

Proctoring and Disciplinary Actions To ensure integrity, the system continuously records proctoring actions, such as keystrokes, focus changes, and clipboard events, during an active exam attempt. The system compares different attempts to calculate similarity scores and automatically generates flags for suspicious behavior. If an attempt is flagged, an instructor conducts a disciplinary review. This review may result in a formal sanction against the student. If a sanction is issued, the student has the right to file an appeal, which is subsequently reviewed by an administrator or instructor for a final decision.